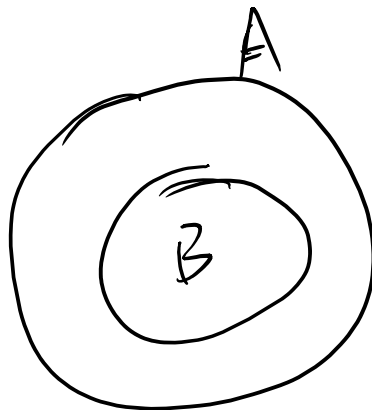


$A = \text{audio player}$

$B = \text{compact disc player.}$

$$B \subseteq A$$



$$B \cap A = B$$

$$P(A/B) = P(A)$$

This says that: Probability of A does not change once B is known

A = heads on tossing a coin

$$P(\text{heads on tossing a coin}) = 0.5$$

$$P(\text{heads on tossing a coin / every day}) = 0.5$$

Definition of Conditional independence

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Since $P(A|B) = P(A)$ [A is independent of B].

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A) = \frac{P(A \cap B)}{P(B)}$$

$$\boxed{P(A \cap B) = P(A)P(B)}$$

$$P(L \text{ from first bag, } L \text{ from second bag}) \\ = P(L \text{ from first bag}) P(L \text{ from second bag})$$

because draws from both bags are independent

$$= \left(\frac{1}{5}\right) \left(\frac{1}{6}\right) = \frac{1}{30}$$

At least one right handed

R = right handed

L = left handed

RLL, RLC, RLR, LRL, LRR

$$\left. \begin{array}{l} P(RLL) \\ + \\ P(RLR) \\ + \\ P(RRL) \end{array} \right\}$$

but this is time consuming

enumerating 7 combinations is time

consuming

$$\underbrace{P(RLL) + P(RLR) + \dots}_{\text{consuming}} = \frac{1 - P(LL)}{1 - \left(\frac{28}{300}\right)^3}$$

Meaning of $A_1, A_2, \dots, A_K$ being
mutually exclusive and exhaustive?

mutually exclusive $\rightarrow$ no two of these
events can occur together.

exhaustive $\rightarrow$ nothing outside of $A_1, A_2, \dots, A_K$

can occur

$$A_1 \cup A_2 \dots \cup A_K = S$$

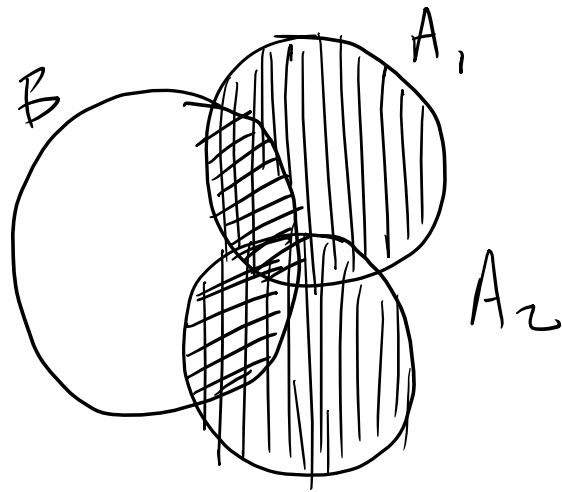
$$B = B \cap S$$

$$= B \cap (A_1 \cup A_2 \dots A_k)$$

By distributive law

$$B \cap (A_1 \cup A_2 \dots \cup A_k) = (B \cap A_1) \cup (B \cap A_2) \dots \cup (B \cap A_k)$$

Why is the distinctive law true?

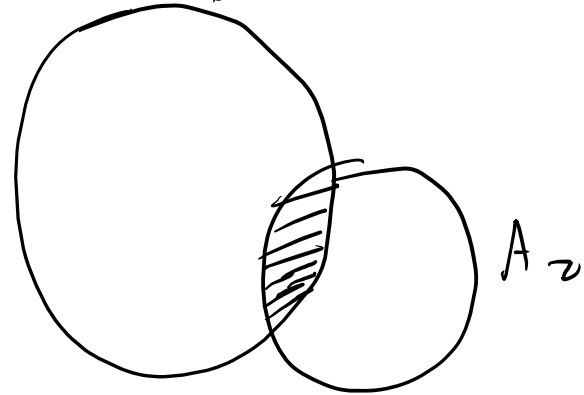
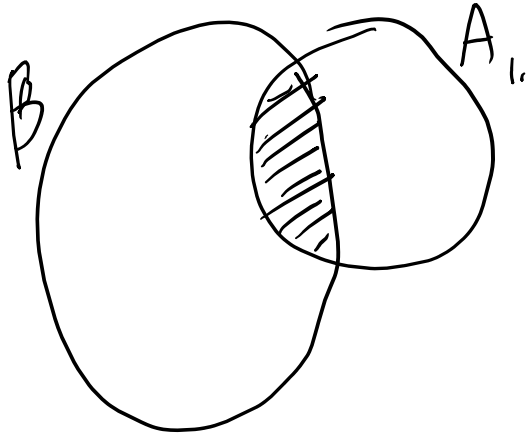


$$B \cap (A_1 \cup A_2)$$

$A_1 \cup A_2$ is area
that is vertically
hatched

$$B \cap (A_1 \cup A_2) = \text{both vertically \& horizontally hatched}$$

$$B \cap (A_1 \cup A_2) \stackrel{?}{=} (B \cap A_1) \cup (B \cap A_2)$$

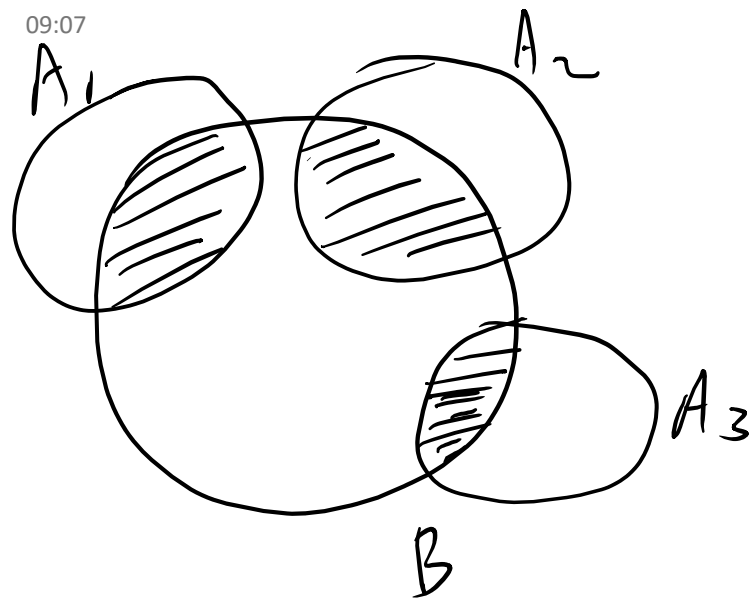


General proof technique is such problems

To show that $\text{set } M = \text{set } N$

- ① take any element in M and show it also belongs to $N \rightarrow M \subseteq N$
- ② take any element in N and show it belongs to $M \rightarrow N \subseteq M$

$$\textcircled{1} \ \& \ \textcircled{2} \Rightarrow \underline{N = M}$$

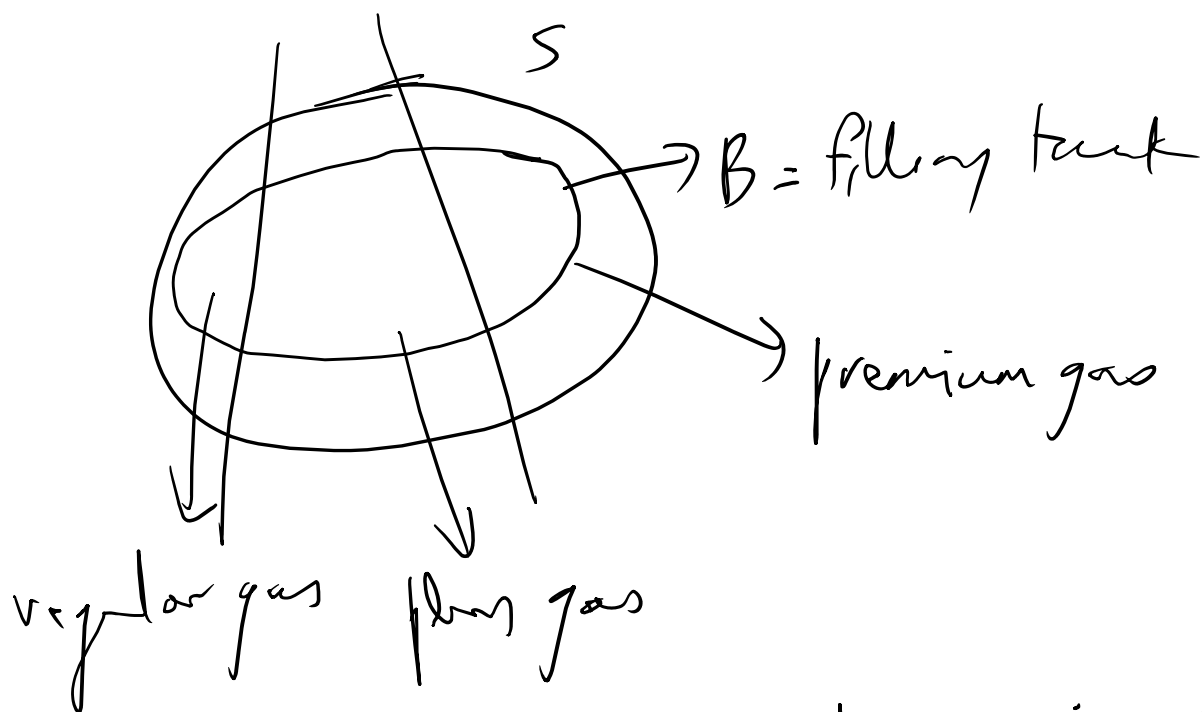


Note: if $P(A) + P(B) = 1$, it does not mean necessarily that A and B are mutually exclusive

[Example: $P(\text{heads on two independent unbiased coins}) = \frac{1}{2}$, but they are not mutually exclusive]

if A and B are mutually exclusive & exhaustive

$$\underline{P(A) + P(B) = 1}$$



Can you describe what outcomes in the sample space look like?

Each outcome will be a tuple
 = (regular gas, filled tank)

(regular gas, did not fill tank)

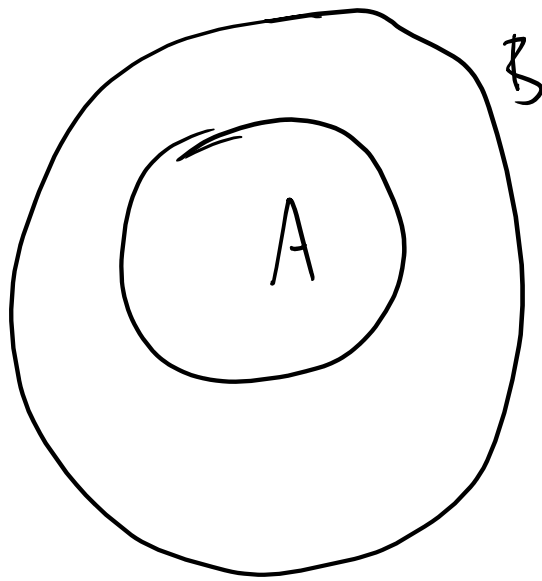
(plus gas, filled tank)

regular gas event = { (regular gas, filled tank),
 (regular gas, did not fill tank) }

1 even

1

1 year / 1 full track



if $A \subseteq B$

$$\underline{A = A \cap B}$$

bonus scheme = B

Managers are X, Y, Z } mutually exclusive

$$B = B \cap X \cup B \cap Y \cup B \cap Z$$

$$P(B) = P(B/X)P(X) + P(B/Y)P(Y) + P(B/Z)P(Z)$$

Let $H_1, H_2, \dots, H_n$ be mutually exclusive
exhaustive hyp-theses.

Why is $\sum_{i=1}^n P(H_i/E) = 1$?

H_i = hyp-thesis i

E = evidence.

$\bigcup_{i=1}^n H_i = \Omega = S$ (sample space).

$$P(H_i | E) = \frac{P(H_i \cap E)}{P(E)} \text{ provided } P(E) > 0$$

$$\sum P(H_i | E) = \frac{\sum_{i=1}^n P(H_i \cap E)}{P(E)}$$

$$= \frac{1}{P(E)} \left(\sum_{i=1}^n P(H_i \cap E) \right)$$

The events $H_i \cap E$ are all mutually exclusive
 Since H_i are mutually exclusive

$$P\left(\bigcup_{i=1}^n H_i \cap E\right) = P(H_1 \cap E) + P(H_2 \cap E) + \dots + P(H_n \cap E)$$

$$\bigcup_{i=1}^n (H_i \cap E) = \left(\bigcup_{i=1}^n H_i\right) \cap E \quad \left[\text{distributive law}\right]$$

$$= \Omega \cap E = E$$

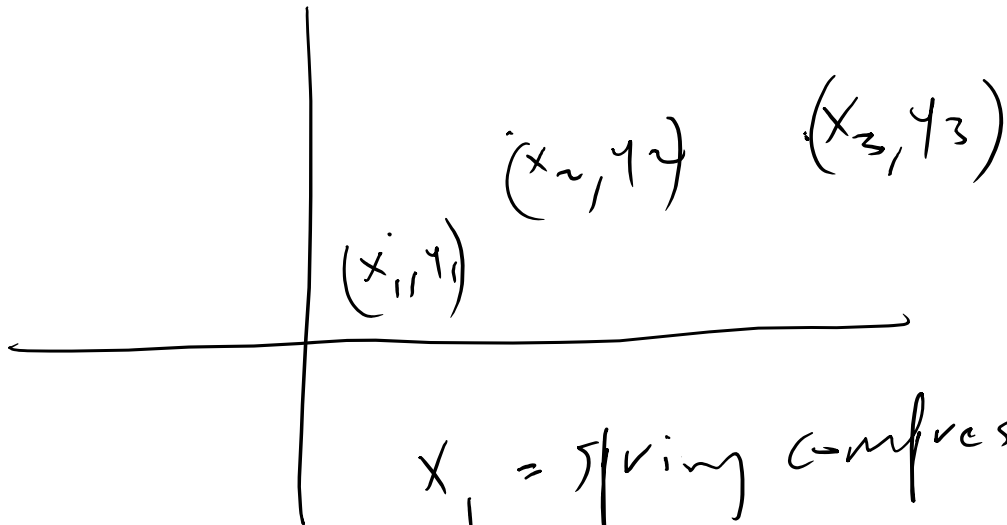
$$P\left(\bigcup_{i=1}^n H_i \cap E\right) = P(E)$$

$$\therefore P(E) = P(H_1 \cap E) + P(H_2 \cap E) + \dots + P(H_n \cap E)$$

$$~~P(E)~~ = ~~P(E)~~ P(H_1/E) + ~~P(E)~~ P(H_2/E) + \dots + ~~P(E)~~ P(H_n/E)$$

$$\sum_{i=1}^n P(H_i/E) = 1$$

What is data here (x_1, y_1)



x_1 = spring compression length

y_1 = amount of force generated by spring

Hypothesis is the natural law that
best explains the data

$$\text{Force of spring} = k (\text{spring compression length})$$

(one hypothesis)

Another hypothesis

$$\text{Force of spring} = k (\text{spring compression length})^2$$

Initially

$$\left. \begin{aligned} P(\text{linear def}) &= 0.3 \\ P(\text{quadratic def}) &= 0.7 \end{aligned} \right\} \text{in the absence of any previous data}$$

$$P(\text{observing given data} / \text{linear def}) = 0.99$$

$$P(\text{observing given data} / \text{quadratic def}) = 0.01$$

The best fit to the given data is
a linear fit $\rightarrow$ follows

"linear def"